Rules

Accumulator & Cell Monitoring

- BMS must monitor individual voltage cells of tractive battery
- Cannot solder wires to cells for voltage monitoring input of BMS
- BMS voltage sensing connections must be protected against short circuits
- BMS must monitor cell or module temperatures of tractive battery

Fault Detection & Safety Functions

- BMS must detect over-voltage, under-voltage, and over-temperature conditions
- BMS must open shutdown circuit if any monitored parameter exceeds safe limits
- BMS must detect an open circuit or failure in voltage monitoring connections
- BMS must remain operational whenever tractive system is active
- BMS must remain operational whenever tractive battery is connected to a charger

Shutdown Circuit Integration

- BMS must be part of shutdown circuit
- BMS must be Normally Open
- BMS must have completely independent circuits to open the shutdown
- BMS must open shutdown circuit in the event of an internal fault or loss of power
- BMS must command opening of Accumulator Isolation Relays via shutdown circuit

Isolation & Architecture

- BMS must have galvanic isolation between tractive system and low voltage systems
- BMS must have galvanic isolation at each Module-to-Module boundary, as approved in ESF

Design Allowances

- BMS should be capable of detecting an open fusible link and opening shutdown circuit if a fault is detected

Components Needed

Cell Voltage Monitoring

- Voltage sensing connections for each cell/group of cells
- Sense wires that connect to cells without soldering
- Resistors or fuses on each sense wire
- Hardware to detect if a voltage sense wire is disconnected

Temperature Monitoring

- Temperature sensors on cells or modules and wiring to connect them to BMS
- Circuitry to read temperature signals

BMS Control Unit

- Microcontroller to control BMS
- Internal fault detection such as reset circuit

Shutdown Circuit Interface

- Output that connects BMS to the shutdown circuit
- 2 independent shutdown-opening paths
- Output drivers that can open shutdown loop
- Interface to control Accumulator Isolation Relays through the shutdown circuit

Power Supply

- Low-voltage power supply for BMS
- Fuse for input protection
- Voltage regulators

Galvanic Isolation

- Isolated interfaces for voltage sensing
- Isolated interfaces for shutdown signals
- Isolation barriers between battery modules

Other

- Power/sensing paths for when battery is connected to a charger
- Connectors (rated for battery voltage)
- Insulated connectors for sense wiring